Chapter 1

Overview of surface wave methods

Since Lord Rayleigh predicted their existence (Rayleigh 1885), surface waves have attracted the interest of an increasing number of researchers embracing disciplines as diverse as solid-state physics, microwave engineering, geotechnical engineering, nondestructive testing, seismology, geophysics, material science, ultrasonic acoustics, and others. Despite their marked differences, these disciplines share the goal of exploiting surface waves propagating along the boundary of a domain to obtain information about the interior of that domain, usually expressed in terms of one or more scalar fields.

Surface waves are appealing because they are ideal for the development of noninvasive techniques for material characterization from a very small scale, less than a millimeter (e.g., ultrasonic surface waves used to identify material defects), to a very large scale, more than a kilometer (e.g., earthquake-generated surface waves used to investigate the structure of the Earth's crust and upper mantle). At an intermediate scale, geophysicists and geotechnical engineers use surface waves for the characterization of geomaterials. The fundamental idea is the same for all these applications: to use the geometric dispersion of surface waves to infer the relevant medium properties by solving an inverse problem for parameter identification.

The present chapter is devoted to an introduction to surface wave methods. It describes their basic principles and historical developments, with references to geotechnical and geophysical near-surface characterization as well as deep exploration and exploration seismology. The most common applications related to near-surface characterization will be discussed, and some crucial aspects related to the application of the technique will be examined to provide guidance to new users and to identify elements for further research on the topic. Finally, the advantages and limitations of surface waves relative to other seismic methods will be discussed, pointing out the significance and relevance of surface wave tests in the context of engineering site characterization.

Although this chapter mainly focuses on Rayleigh waves, as they are more widely used for characterization purposes, the same concepts apply to the use of Love and Scholte waves for site characterization as discussed in Chapter 8.

1.1 SEISMIC WAVES

Soils and rocks are complex, multiphase, particulate and discontinuous materials; thus, their mechanical behavior cannot in general be described using simple models. Soils exhibit marked nonlinear and irreversible behavior, starting from the very initial stage of loading. Although soils are particulate materials, continuum mechanics approaches are frequently used, accounting for the role of fluid in saturated media through the effective stress concept introduced by von Terzaghi (1936). The dynamic behavior of soil is very complex and depends on a variety of factors. Among these, the dependency on the strain level is of primary interest with respect to the focus of this book. For very small strains, soils exhibit an almost linear stress–strain relationship, so that the assumption of a linear elastic constitutive model is reasonable. Outside the small-strain region, the behavior is far more complex, and its description requires the adoption of advanced constitutive models, which is outside the scope of this book.

The strain level associated with geophysical testing is typically very small, so it is widely accepted that linear elastic theory provides a consistent framework for the interpretation of seismic tests. Linear elastic models can be used directly only for a restricted number of boundary value problems of soil dynamics (e.g., ground vibrations predictions). However, small-strain parameters play a significant and relevant role for a wide range of geotechnical problems (Burland 1989; Atkinson 2000).

Equations of motion for a linear elastic solid will be presented in detail in Chapter 2. The effect of a sharply applied, localized disturbance in a physical medium rapidly spreads in space; this is commonly addressed as wave propagation (Graff 1975). Two different types of body (or bulk) waves propagate in an unbounded, homogeneous, and linear elastic medium: P-waves (primary or compressional or longitudinal waves) and S-waves (secondary or shear or distortional or equivoluminal waves). P-waves propagate with particle motion in the same direction of the propagation and cause volume change without distortion. They propagate at a velocity greater than S-waves, for which the particle motion is perpendicular to the direction of propagation.

The velocities of propagation of seismic waves in a linear elastic solid are associated with the medium's mechanical parameters through simple relationships. Indeed, the mechanical response of an elastic medium is fully characterized by two elastic constants, for example, Lamé's constants λ and μ (the latter being the shear modulus G in engineering notation). Seismic wave

velocities can be expressed with simple relationships between such constants and the material mass density ρ (see Section 2.1.3)

$$\begin{cases} V_P = \sqrt{\dfrac{\lambda + 2G}{\rho}} \\ V_S = \sqrt{\dfrac{G}{\rho}} \end{cases} \tag{1.1}$$

The velocity of propagation of body waves is directly linked to the stiffness of the medium and is not frequency dependent in linear elastic materials. In particular, the velocity of propagation of P-waves is associated with the (small-strain) longitudinal modulus, whereas the velocity of propagation of S-waves is associated with the (small-strain) shear modulus. Equation 1.1 forms the basis for the use of seismic waves in material characterization, showing that the elastic constants can be easily determined if the seismic wave velocities are measured experimentally.

P-wave velocities (Equation 1.1) are of limited value in saturated soils because of the role played by the pore fluid in determining the overall response of the soil. This can be explained in detail via Biot's theory of wave propagation in saturated porous media (Biot 1956a, 1956b). It can be shown that the velocity of propagation of P-waves is strongly influenced by the compressibility of the pore fluid rather than the soil skeleton, whereas the influence of the pore fluid on S-wave propagation is negligible and is only linked to the change in mass density because the pore fluid has no shearing resistance. Thus, S-waves (Equation 1.1) are widely used for soil characterization because their velocity is directly related to the shear modulus of the soil skeleton.

Finally, although linear elasticity is adequate for modeling the stiffness of soils at small-strain levels, it is not able to describe the energy dissipation in loading–unloading cycles that is observed in soils even at very small strains. From this point of view, linear viscoelasticity provides a more consistent framework for the interpretation of seismic tests. Wave propagation in linear viscoelastic media can be modeled using the elastic–viscoelastic correspondence theorem as discussed in Section 2.5.

1.1.1 Seismic tests for site characterization

The close link between the velocity of propagation of seismic waves and the elastic constants of the medium makes seismic waves useful for geotechnical site characterization. Several tests have been designed to measure or estimate seismic wave velocities *in situ* with the primary purpose of assessing the variation with depth of the shear wave velocity (i.e., a shear wave velocity profile). In particular, the great advantage of testing geomaterials in their undisturbed state (especially important for hard-to-sample soils) and

the large volume of material involved are two main advantages of *in situ* seismic methods. Other applications are discussed by Jamiolkowski (2012). For example, seismic wave velocities can be used to evaluate soil porosity on the basis of Biot's theory (Foti et al. 2002). Although the most common application is the measurement of seismic wave velocities, engineers and seismologists have used *in situ* seismic methods to measure the attenuation of waves and to compute the material damping ratio of soils.

Seismic methods for site characterization are often divided into two broad categories: *invasive* tests and *noninvasive* tests. The former methods require a borehole (cross-hole, down-hole, up-hole, P-S suspension logging) or the insertion of a probe in the soil (seismic cone penetration test or seismic dilatometer), the latter methods are entirely conducted on the ground surface (seismic reflection, seismic refraction, and surface wave tests). In general, the noninvasive tests are affected by a larger degree of uncertainty, but they allow the exploration of larger, more representative volumes, and they are usually more cost-effective than invasive tests.

In cross-hole tests (CHTs), source and receiver(s) are placed at the same depth, and the wave velocity is calculated on the basis of the travel time. Being based on a direct measurement, CHTs are usually considered the most reliable and accurate method. Nevertheless, the need for two or three cased boreholes increases the cost of the test.

In down-hole tests, seismic waves are generated on the ground surface, and the travel times are measured in a borehole at different depths using one or more receivers, which are progressively moved to deeper positions. The interpretation can be based on the trends of travel times with depth or on the travel-time interval between the arrivals at two receivers. Appealing alternatives in geotechnical site characterization are based on receivers mounted within cone penetration test (CPT) or dilatometer test (DMT) probes, which allow the execution of down-hole measurements at different stages during the insertion of the probe in the soil with no need for a dedicated borehole.

In P–S suspension logging, a single, cable-wired instrument is inserted in the borehole. The instrument contains a seismic source and two or more receivers and provides a local measurement of seismic wave propagation at different depths. The method is unique in that the instrument is suspended in a fluid-filled borehole, with the fluid providing the mechanical coupling between the instrument and the surrounding soil. This approach is ideal for measurements in very deep boreholes, and it is widely used in the oil and gas industry, although its use for engineering applications is growing.

Noninvasive seismic techniques include seismic reflection and refraction methods. Both methods have been developed using P-waves but are currently used also with S-waves, although there are some difficulties in generating high-energy, horizontally polarized shear waves.

In seismic refraction methods, soil characterization is based on arrival time on the ground surface of seismic waves that have been critically refracted at

interfaces among layers having a different velocity or continuously refracted in an inhomogeneous medium. They suffer from intrinsic limitations related to the presence of velocity inversions or hidden layers (Sheriff and Geldart 1995; Reynolds 1997), which lead to incorrect estimation of interface depth and seismic velocities.

Seismic reflection surveys are primarily aimed at detecting and imaging interfaces among different layers. They require complex processing and interpretation procedures. Although some high-resolution applications for near-surface characterization have been proposed, these methods remain mainly devoted to deep exploration.

Surface wave methods are noninvasive methods based on the solution of the inverse problem of Rayleigh wave propagation, which is aimed at estimating the shear wave velocity profile of the subsurface. Rayleigh waves are easily generated and detected at the ground surface, providing a powerful tool for site characterization.

1.2 SURFACE WAVES

Several of the properties of surface waves make them particularly well suited for geomaterial characterization. They originate from the condition of vanishing stress at a boundary of a domain (e.g., the surface of the Earth), and their radiation pattern (see Figure 1.1) is essentially two-dimensional (2D)

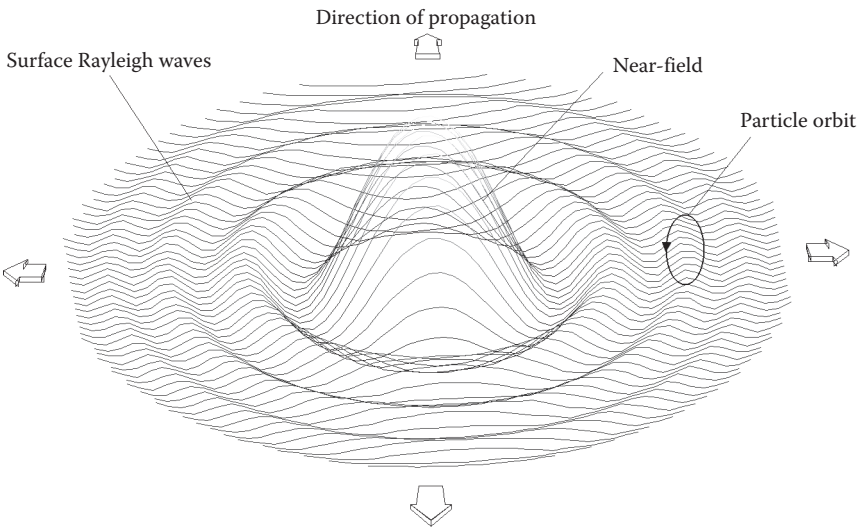

Figure 1.1 2D radiation pattern of Rayleigh surface waves generated by a vertical point source.

and thereby characterized by a much lower rate of geometric attenuation than body (or bulk) waves whose energy spreads in horizontal and vertical directions. As an example, Rayleigh surface waves generated by a line source in a homogeneous elastic half-space do not exhibit any geometric attenuation, whereas for a point load, the rate of spatial decay is proportional to the inverse of the square root of the distance from the source. Conversely, for a point source, the geometric attenuation factor of body waves propagating along the boundary of an elastic half-space is proportional to the inverse of the square of the distance (Ewing et al. 1957). Thus, at distances on the order of one to two wavelengths from the source, the contribution of body waves becomes negligible, and the wave field is dominated by Rayleigh waves. Lamb (1904) was among the first scientists to recognize this fundamental property of surface waves and its implications for the transmission of earthquake energy at large distances.

In the direction orthogonal to that of propagation, the displacement field generated by a surface wave decays exponentially because no energy is propagated in the interior of the half-space. It can be shown that most of the strain energy associated with surface wave motion is confined within a depth of about one wavelength λ from the free boundary (Achenbach 1984). Hence, Rayleigh waves with long wavelengths penetrate deep into the interior of a medium. Because wavelengths are proportional to the inverse of frequency in harmonic waves, this statement can be interpreted as follows: high-frequency waves are confined to shallow depths within the medium, whereas low-frequency components involve motion also at large depths.

1.2.1 Geometric dispersion

Rayleigh waves in a homogeneous, isotropic, linear elastic half-space are not dispersive; that is, their velocity of propagation is a function of the mechanical properties of the medium but not a function of frequency. In vertically heterogeneous media, the phenomenon of geometric dispersion arises, which results in the phase velocity of Rayleigh waves being frequency dependent. The dispersive nature of Rayleigh waves propagating in a vertically heterogeneous medium forms the basis of surface wave testing. Consider the example shown in Figure 1.2, with a horizontally layered medium consisting of two layers overlying a half-space. On the left, the approximate vertical particle motion associated with a high-frequency (i.e., short wavelength) Rayleigh wave is shown. Most of the particle motion is confined to within about one wavelength from the free surface. In this case, the particle motion occurs almost exclusively in Layer 1. Hence, the material properties of Layer 1 control the velocity of the Rayleigh wave. The right side of the figure illustrates the vertical particle motion associated with a low-frequency (i.e., long-wavelength) Rayleigh wave. In this case,

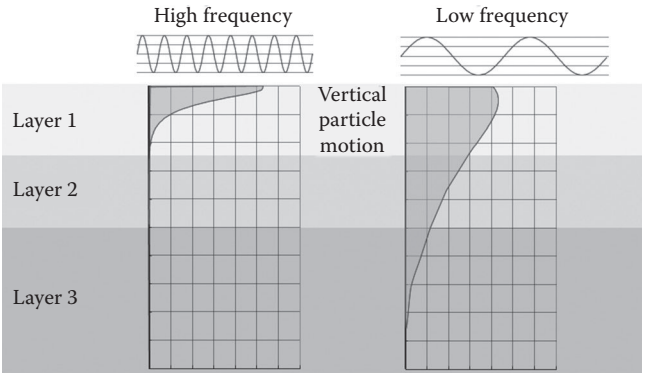

Figure 1.2 Geometric dispersion of Rayleigh waves: trends with depth of the vertical particle motion associated with the propagation of two harmonic waves in a layered medium.

the particle motion extends to a greater depth, and there is significant particle motion in Layers 1 and 2 and less in Layer 3. The velocity of this low-frequency Rayleigh wave is controlled by some combination of the material properties of all three layers, perhaps in rough proportion to the relative amount of particle motion occurring within each layer.

The dispersive nature of Rayleigh waves can be used for identification purposes; by experimentally measuring the dispersion curve (i.e., the variation of Rayleigh phase velocity with frequency) that is associated with a given site, it is possible, via an inversion process, to determine the shear wave velocity profile of the site. More broadly, the objective is to characterize the interior of a medium from measurements of a 2D surface wave field at its free boundary. Mathematical formalization of this objective leads to the theory of *inverse problems*, specifically to a particular branch of this theory known as *system* or *parameter identification problems* (Engl 1993). Solution and properties of such problems depend strongly upon the constitutive model used to describe the response of a medium to the propagation of disturbances, as well as on the properties of its boundaries.

To summarize the concept behind the use of geometric dispersion for site characterization, assume that the stratified medium in Figure 1.3a is characterized by increasing stiffness with depth, so that the shear wave velocity of the top layer is less than the velocity of the second layer, which in turn is less than the velocity of the half-space below. In this situation, a high-frequency Rayleigh wave (Figure 1.3b), traveling in the top layer, will have a velocity of propagation slightly lower than the shear wave velocity of the first layer. On the contrary, a low-frequency wave (Figure 1.3c) will travel at a higher velocity because it is influenced by the underlying

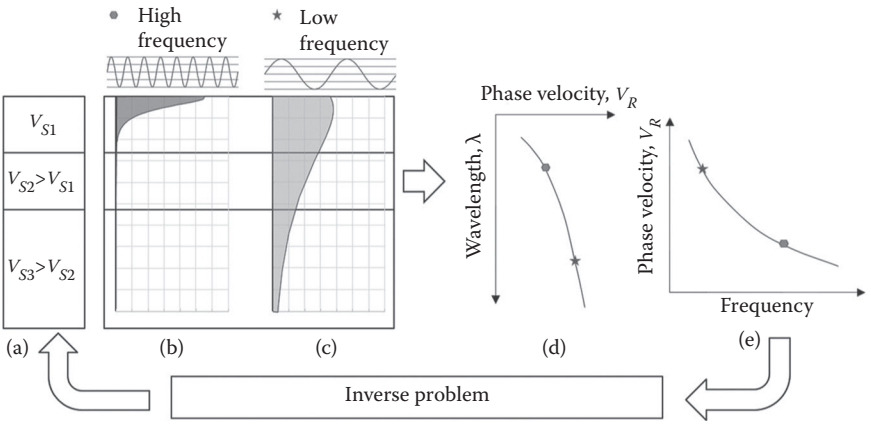

Figure 1.3 Parameter identification on the basis of geometric dispersion.

stiffer materials as well. This concept can be extended to other frequency components. A plot of phase velocity versus wavelength (Figure 1.3d) will hence show an increasing trend for longer wavelengths. Considering the intimate relationship between wavelength and frequency, this information can also be represented in a plot of phase velocity versus frequency (Figure 1.3e), which is commonly called a dispersion curve. This example shows that, for a vertically heterogeneous medium, the dispersion curve contains information about the variation of medium parameters with depth. This is the so-called *forward problem*, and it will be presented in detail in Chapter 2. If the dispersion curve is obtained experimentally (see Chapter 4), it is then possible to solve the *inverse problem* to obtain the shear wave velocity profile (i.e., the medium parameters are identified on the basis of the experimental data collected on the boundary of the medium; see Chapter 6). This is the essence of surface wave methods; additional details are presented here and in subsequent chapters.

1.3 TEST METHODOLOGY

Most surface wave tests are aimed at measuring the shear wave velocity profile, which is closely linked to the vertical variation of the small-strain shear modulus. Usually, this task is accomplished by first obtaining an experimental dispersion curve from measured field data. These data are subsequently used for the solution of the parameter identification problem. This latter step implies the choice of a reference model for the interpretation, which in most cases is a one-dimensional (1D) model comprising a stack of homogeneous, linear elastic layers.

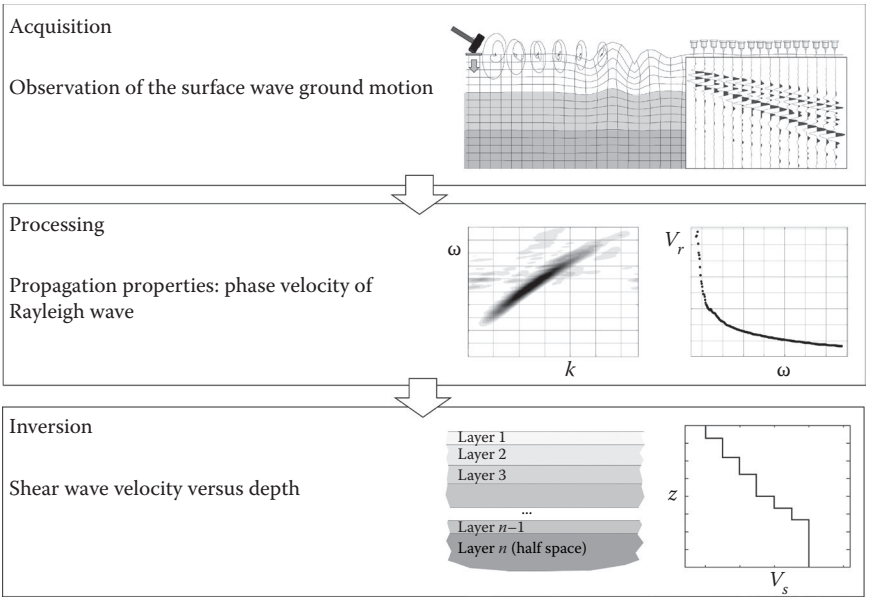

Figure 1.4 Flow chart of surface wave tests.

Figure 1.4 shows the standard procedure for surface wave tests, which can be subdivided into three main steps:

1. Acquisition of experimental data
2. Signal processing to obtain the experimental dispersion curve
3. Inversion process to estimate site properties

It is important to recognize that the individual steps are strongly interrelated and their interaction must be adequately accounted for during the whole interpretation process. Many types of surface wave tests have been developed for near-surface site characterization, often addressed using a wide range of acronyms. They may differ in any one of these steps regarding the type of source and the number of receivers, the signal processing technique, or the inversion strategy and algorithm, but the overall principle is essentially the same.

Surface wave tests are typically performed using Rayleigh waves because of the simplicity of generating and detecting them on the ground surface. Similar applications using Love waves, which originate from horizontally polarized surface waves, are also possible, but they suffer from some limitations related to layer stratigraphy and to the difficulties in generating Love waves. Applications at the seafloor are also becoming popular. They employ

the same principles but use Scholte waves, which are surface waves that propagate along the interface between a fluid layer and a solid half-space (see Sections 1.6.4 and 8.2).

Each step involved in surface wave testing is briefly discussed in the following sections, whereas the reader is referred to the subsequent chapters in the book for additional details and technical aspects.

1.3.1 Acquisition

Surface wave data are typically collected using a variable number of receivers at the ground surface, deployed according to several possible geometrical configurations. Variations can be introduced in the choice of receivers and acquisition device and in the generation of the wave fields.

The receivers adopted in engineering and environmental near-surface applications, and in conventional exploration surveys, are typically geophones (i.e., electrodynamic velocity transducers). Geophones are normally preferred for their high sensitivity, robustness, and relatively low cost. Low-frequency geophones (with natural frequencies less than 2 Hz) tend to be bulky, less robust, more expensive, and operationally less efficient. A high-sensitivity microelectromechanical systems (MEMS) accelerometer can be beneficial for low-frequency surface wave acquisition. In very small-scale applications, e.g., the characterization of pavement systems, accelerometers are often used because of the need for high-frequency, high-amplitude signals.

Several types of devices can be used for the acquisition and storage of signals. Basically, any hardware having an A/D converter and the capability to store digital data can be adopted, ranging from seismographs to dynamic signal analyzers to systems comprising data acquisition boards connected to personal computers (PCs) or laptops. Commercial seismographs for geophysical prospecting are typically the first choice because they have high-end specifications and they are designed to be used in the field. Thus, they are robust, waterproof, and resistant to dust. Modern seismographs are composed of scalable acquisition components used in conjunction with field computers, allowing preliminary processing of data on-site.

For the generation of Rayleigh surface waves, several different types of sources can be used, provided they have sufficient energy in the frequency range of interest for the specific application. Transient, impact sources are often preferred because they are inexpensive and rugged. They range from small hammers for generating high-frequency surface waves to large falling weights, which generate more low-frequency components. Seismic guns may be used for shallow applications, although explosives are usually limited to deep exploration because of logistic issues. Appealing alternatives are continuous sources, which are able to generate

controlled, harmonic waves. Also in this case, the size of the source varies from relatively small, electromagnetic shakers to large, truck-mounted Vibroseis™. The drawback of such sources is essentially their cost and the need for long acquisition times on site when light sources are used with monochromatic signals.

It is also possible to use passively generated surface waves. In this case, the need for an "active" transient or continuous source is avoided by recording "passive" ambient noise, often called microtremors. Microtremors include cultural noise generated by human activities (traffic on highways, construction activities, and so on) and vibrations arising from natural events. A great advantage is that microtremors are usually rich in low-frequency components. Hence, passive surveys provide useful information for deep characterization (tens or hundreds of meters). However, the level of detail close to the ground surface is typically low because microtremors lack high-frequency energy. This limitation can be overcome by combining active and passive measurements (Figure 1.5). Field acquisition is discussed thoroughly in Chapter 3.

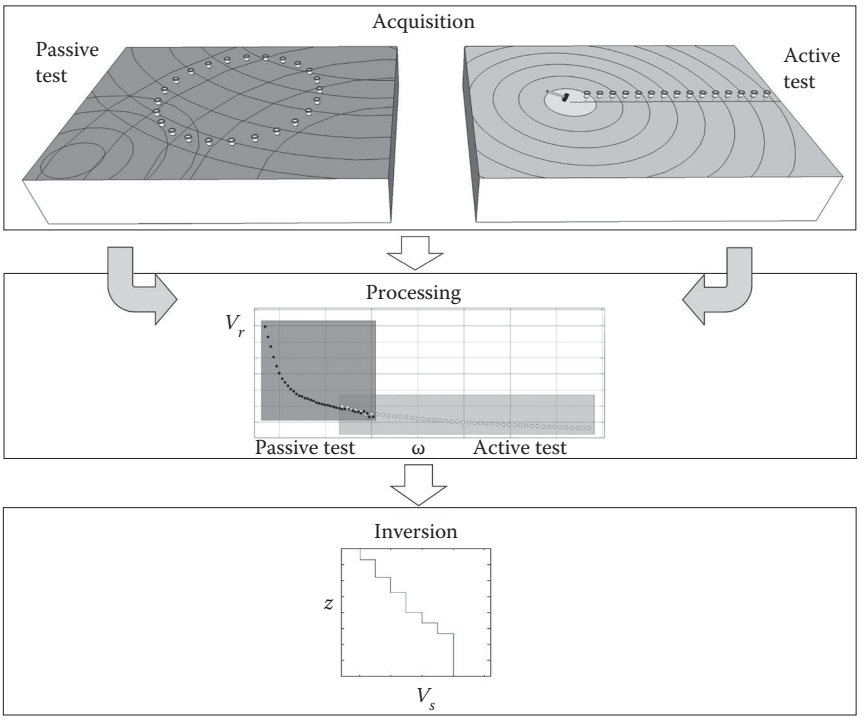

Figure 1.5 Combination of active-source and passive-source surface wave measurements.

1.3.2 Processing

The field data are processed to estimate the surface wave propagation parameters, typically the experimental dispersion curve. Different surface wave test procedures apply a variety of signal processing techniques, but they mainly rely on the Fourier transform to decompose the time history of vertical particle motion into its frequency components. Indeed, using Fourier analysis, it is possible to separate the different frequency components of a signal, which are subsequently processed to estimate phase velocity using different approaches that are a function of the test configuration and the number of receivers. In the example provided in Figure 1.6, the dispersion curve is obtained by selecting the peak amplitudes of the frequency–wavenumber spectrum. Specific processing techniques will be discussed in detail in Chapter 4.

Some types of equipment (e.g., PC-based systems) allow for processing the experimental data directly in the field. Often, the simple visual screening of the recorded seismic traces is not sufficient; surface wave components

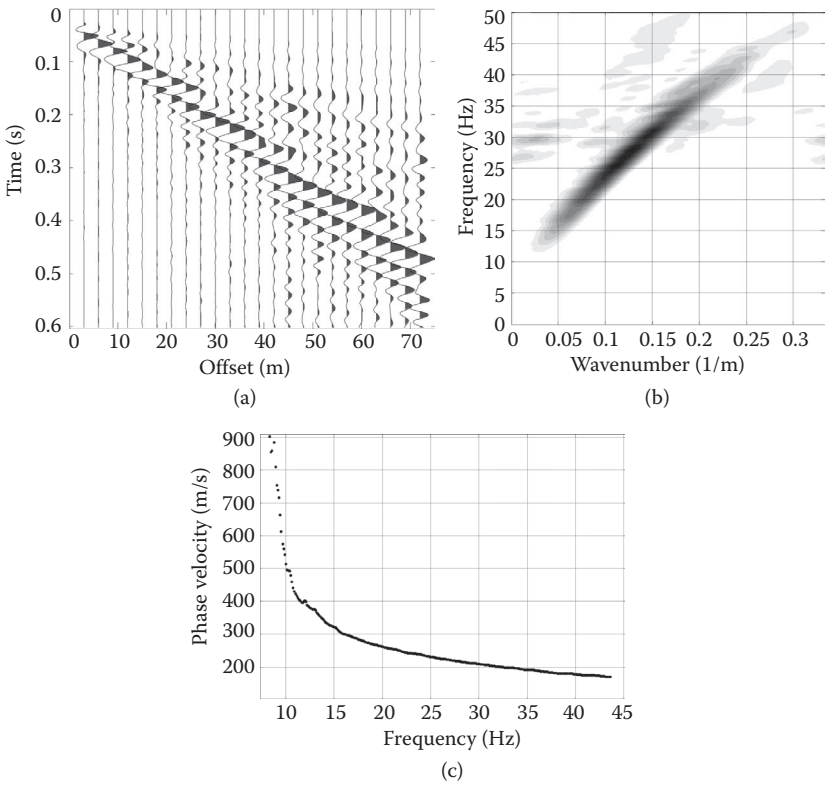

Figure 1.6 Example of processing of experimental data using the frequency–wavenumber analysis: (a) field data; (b) *f–k* spectrum; (c) dispersion curve.

are grouped together, and without signal processing it is not possible to judge the quality of the data over the desired frequency or wavelength range. An assessment of the frequency range with adequate signal quality can be particularly useful to assess the need to change the acquisition setup or to gather additional experimental data.

There are several test variants in which the experimental dispersion curve is replaced by other types of data. These include the inversion of full, time-domain waveforms (e.g., Tran and Hiltunen 2012) and the inversion of the Fourier frequency spectra of observed ground motion (Szelwis and Behle 1987). Such strategies are not commonly used in practice, especially in near-surface site characterization. Moreover, the experimental dispersion curve is informative about trends to be expected in the inverted shear wave velocity profile, so its visual inspection is important for the qualitative validation of the results.

Surface wave data can also be used to characterize the dissipative behavior of soils. The spatial attenuation of surface waves is associated with the internal dissipation of energy. Using a procedure substantially analogous to the one outlined in Figure 1.4, it is possible to extract from field data the experimental *attenuation curve*, that is, the coefficient of attenuation of surface wave as a function of frequency, and then use these data in an inversion process aimed at estimating the material damping ratio (or the quality factor) profile for the site (see Chapters 5 and 6). Furthermore, the procedure for attenuation and dispersion analysis and inversion can be coupled leading to simultaneous determination of shear modulus and material damping ratio profiles (see Chapter 6).

1.3.3 Inversion

The solution of the Rayleigh inverse problem is the final step in test interpretation. The solution of the forward problem (Chapter 2) forms the basis of any inversion strategy. Assuming a model for the soil deposit, model parameters are identified by minimizing an objective function representing the distance between the experimental and the theoretical dispersion curves. The objective function can be expressed in terms of any mathematical norm (usually the root mean square or RMS) of the difference between experimental and theoretical data points. In practice, the set of model parameters that produces a solution of the forward problem (a theoretical dispersion curve) as close as possible to the experimental data (the experimental dispersion curve of the site) is selected as solution of the inverse problem (e.g., Figure 1.7).

This objective can be reached using a variety of strategies. A major distinction arises between local search methods (LSMs), which start from a tentative shear wave velocity profile and search in its vicinity, and global search methods (GSMs), which attempt to explore the entire space of possible solutions. LSMs are undoubtedly faster because they require a limited number

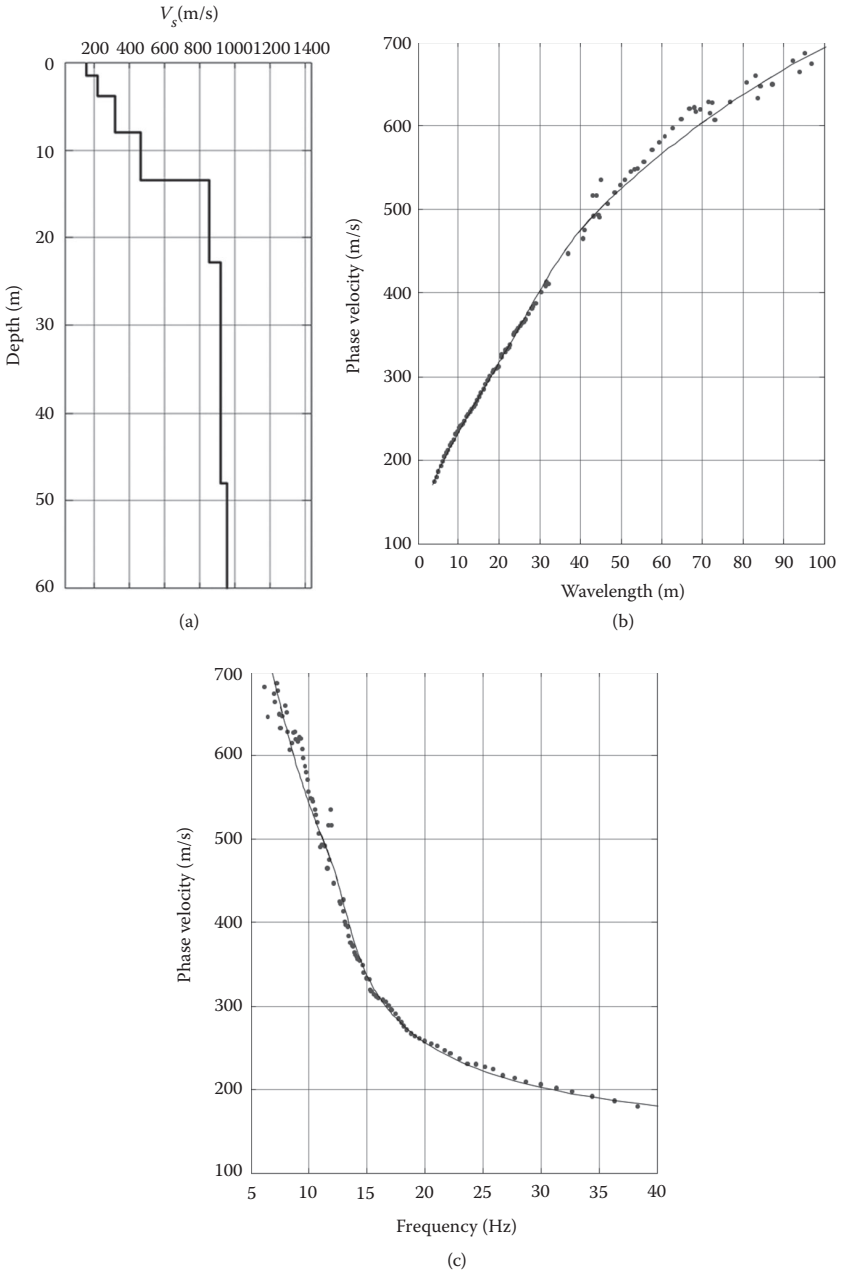

Figure 1.7 Example of inversion process: (a) shear wave velocity profile; (b) comparison between the theoretical and experimental dispersion curves in wavelength–phase velocity domain and (c) in frequency–phase velocity domain.

of runs of the forward problem of Rayleigh wave propagation. However, because a solution is sought in the vicinity of a tentative profile, there is the risk of being trapped in local minima. Conversely, GSMs require substantial computational effort because a large number of runs of the forward problem is required, so that the approach may be quite time consuming.

In general, inverse problems are inherently ill-posed, and a unique solution does not exist. A major consequence is the so-called equivalence problem (i.e., several shear wave velocity profiles may be equivalent in the sense that the theoretical dispersion curves associated with these profiles are at the same distance from the experimental dispersion curve). A meaningful evaluation of equivalent profiles must take into account the uncertainties in the experimental data. Additional constraints and a priori information from borehole logs or other geophysical tests are useful elements to mitigate the equivalence problem.

Infact, engineering judgment plays an important role in the entire test procedure, and the results of fully automated interpretation methods should be carefully examined, with special attention to intermediate results during each step of the test procedure. Knowledge of the theoretical aspects of wave propagation, signal processing, and inversion, and experience are hence essential ingredients for the successful application of surface wave tests.

1.4 HISTORICAL PERSPECTIVE

The interpretation of surface wave data requires the availability of digital records and many computationally intensive tasks. It is therefore not surprising that most advancements in surface wave testing and their widespread application are closely linked to the historical progress in electronics, for data acquisition devices and computers. In particular, the advent of reasonably priced data collection devices and PCs has led to their use in near-surface geophysical and engineering site characterization. Most of the tools for the analysis of seismic records and for the solution of the forward and inverse Rayleigh problems come from the field of seismology and have been successively adapted for engineering applications. Wave field transforms and multichannel data processing inherited contributions from seismic data processing for oil and gas exploration. Nevertheless, the spread of surface wave methods in near-surface applications has led to a number of subsequent developments, which take into account the different scale of the applications.

1.4.1 Global seismology

Energy released by earthquakes travels to a teleseismic distance mainly in the form of long-period surface waves, which represent by far the largest component of seismic records at great distance from the epicenter. Surface waves

have been studied in seismology for the characterization of Earth's interior since the 1920s, but their widespread use started during the 1950s and 1960s thanks to the increased possibilities of numerical analysis and improvements in instrumentation for recording seismic events (Ben-Menhaem 1995; Aki and Richards 2002). In particular, the growth of long-period and broadband seismic networks, started in the 1970s, has led to recent large-scale and global studies on upper mantle structure (Romanowicz 2002).

Most seismological applications have been traditionally based on the analysis of Rayleigh wave components because they are easily identified in the vertical component of seismograms, whereas only with the advent of digital records that allow for a clearer identification of the transverse motion Love wave components have begun to be widely used.

The propagation characteristics of surface waves contain relevant information about the structure of the Earth along the source-station path and the characteristic of the earthquake source. The separation of these two effects is a major challenge for seismologists (Romanowicz 2002). Most crustal and mantle studies use a two-step procedure consisting of the evaluation of surface wave velocity dispersion followed by inversion. The availability of an increasing number of measurement stations has led to different approaches in the evaluation of surface wave velocity.

Originally, the evaluation of group velocity was based on the time interval between peaks and troughs in a single dispersed wave train, while the phase velocity determination was based on the analysis of its Fourier spectrum (Romanowicz 2002). This raised some issues related to the necessity of separating source and path effects. In order to address this issue, many studies began using a "two-station method," in which the dispersion was measured between two recording stations approximately aligned with the epicenter, eliminating the common source phase. The next step was the use of multiple stations spanning a geologically homogenous region (Dziewonski and Hales 1972).

An important problem to be solved was related to the influence of higher modes of propagation in the recorded signals. This motivated the development of several sophisticated filtering techniques based on group velocity features, aimed at separating modal components in teleseismic signals (Dziewonski et al. 1969; Levshin et al. 1994). The inversion of higher modes, whenever they can be extracted from the recording, is advantageous because, as shown in Chapter 2, the particle motion associated with higher modes extends to greater depths than that of the fundamental mode.

The attenuation of surface waves has also been extensively studied for the determination of the anelastic structure of the upper mantle. Working in the frequency domain and exploiting the real and imaginary part of the spectrum, Dziewonski and Steim (1982) obtained information related not only to the velocity model but also to the quality factor (dissipation) model with a formulation based on transfer functions (see also Section 5.3).

Approaches based on a time-domain waveform inversion started to be developed at the end of the 1970s with the formulation of a first-order perturbation theory allowing the computation of synthetic seismograms for a reference model and the evaluation of derivatives with respect to the model parameters. The waveform approach permits the evaluation of model parameters directly from seismograms via a single-step procedure, but the necessity of correcting for crustal structure poses some challenges (Romanowicz 2002).

1.4.2 Exploration geophysics

Traditionally, deep geophysical surveying for hydrocarbon exploration and for regional structure identification has employed seismic reflection and seismic refraction techniques. Sources and receivers are typically placed on the ground surface. Although these techniques are based on body wave propagation, seismograms collected for deep exploration surveys exhibit significant surface wave components. In the gather of Figure 1.8, the first 1500 m

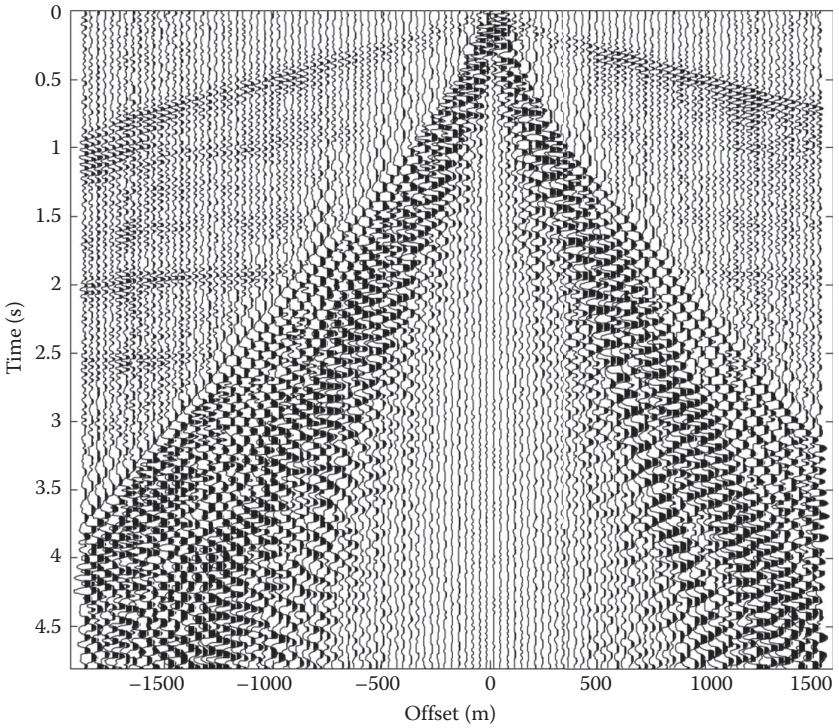

Figure 1.8 Example of common shot gather for seismic reflection surveying. Despite the presence of other events (refracted and reflected P-waves), the gather is dominated by the surface waves.

of offset (i.e., source-to-receiver distance) is dominated by surface waves with multiple modes of propagation.

Surface waves carry a large portion of the energy imparted by shallow sources and they have less geometric attenuation than body waves. Thus, they dominate at small offsets. Moreover, the geometric dispersion in heterogeneous media creates long wave trains with multiple modes that mask reflection signals in the seismograms. For the aforementioned reasons, surface waves are a tedious source of coherent noise in seismic reflection surveys—referred to as "ground-roll." Because of its coherent nature, it is quite difficult to eliminate ground-roll from the shot gathers, and several techniques have been developed to achieve this result (Doyle 1995). As it turns out, the tools used to eliminate ground-roll can be also profitably used to extract information for characterization purposes (Nolet and Pansa 1976; McMechan and Yedlin 1981). An example is shown in Chapter 4 for analyses in the frequency–wavenumber domain.

The interest in the geophysical exploration community toward the exploitation of surface wave components in seismic gathers collected for seismic reflection surveys is increasing as the value of the information that they can provide is recognized (e.g., for the near-surface characterization, for static corrections, or velocity modeling in reflection surveys).

1.4.3 Near-surface applications

Engineering applications of surface wave testing were initially proposed in the 1950s, but they started to become popular and widely used after the introduction of the spectral analysis of surface waves (SASW) method, which for the first time enabled a rapid, theoretically sound test procedure by exploiting the capabilities of new electronic equipment and computers in the late 1970s. Today, the technique is widely used, and the adoption of advanced acquisition and interpretation schemes has made the test even faster and more reliable.

1.4.3.1 Pioneering applications

The first method developed for engineering site characterization was the steady-state Rayleigh method (Section 4.2) proposed by Jones (1958, 1962) and then adopted at the U.S. Army Corps of Engineers Waterways Experiment Station (Ballard 1964). This method was very simple, particularly in the inversion step. Nevertheless, it was ingenious and demonstrated the potential of surface waves for site characterization purposes. The idea came from a series of experiments with mechanical vibrators. The initial application was made using ultrasonic frequencies on concrete slabs to assess their thickness and elastic properties. The success of this technique led to the extension to soil deposits using lower frequencies.

In his field experiments, Jones attempted to use both Rayleigh and Love waves to characterize the soil. The field equipment was composed of a mechanical vibrator and a single receiver. To investigate Rayleigh wave propagation, the vibrator and receiver were placed vertically, while for Love waves the vibrator and receiver were oriented to produce and detect vibrations in a horizontal direction transverse to the testing line.

The experimental dispersion curve was obtained by moving the receiver away from the harmonic source along a straight line, looking for positions where the vibrator and the receiver were in phase. The average distance between two in-phase positions is the wavelength associated with the frequency of the harmonic source. The Rayleigh phase velocity was calculated by multiplying the frequency and observed wavelength. Repeating the process for different frequencies, the dispersion curve was obtained. Once the experimental dispersion curves were obtained, approximate procedures (see Section 6.3) based on theoretical analysis of surface wave propagation were used to infer the shear modulus profile.

1.4.3.2 Spectral analysis of surface waves

The SASW was introduced at the University of Texas at Austin during the late 1970s and early 1980s (Nazarian and Stokoe 1984; Stokoe et al. 1994). Significant advances were introduced with respect to data acquisition/processing and inversion procedures. The acronym itself could in general be used for any surface wave method, but it is associated mainly to this specific technique, because of the large popularity it has achieved.

The SASW test is based on a two-receiver configuration and straightforward signal processing tools (see Section 4.3). The dispersion curve is evaluated by estimating for each frequency component the time delay between the arrivals at the two receivers (expressed by the phase shift) of the wave generated by an active source. Because the two-receiver approach imposes limitations on the frequency range over which the test is effective, the experimental dispersion curve at a site is estimated using several receiver spacings. The individual experimental dispersion curves from different spacings are then combined to obtain a single, composite curve to be used in the inversion process. This procedure may be time consuming and the production rate in the field is adversely affected by the required changes in receiver positions.

Moreover, the interpretation of two-station measurements requires a certain amount of engineering judgment and cannot be easily automated because the periodicity of the phase shift between receivers leads to the necessity of "unwrapping" the phase of the cross-power spectrum used to calculate the Rayleigh phase velocity. The most important limitations are related to the effect of incoherent and coherent noise. The distortions of the experimental local phase induced by the presence of body waves, near-field

effects, lateral variations, and, above all, higher modes can be particularly critical. Despite these limitations, the SASW method occupies a significant place in the historical development of surface wave methods for engineering site characterization and is still in widespread use today.

1.4.3.3 Multistation approaches

The use of multiple receivers enhances the production rate in the field and makes the processing of the data faster, less subjective, and more robust. Early applications of multistation surface wave tests for near-surface characterization were developed in the 1980s (McMechan and Yedlin 1981; Gabriels et al. 1987), but their widespread adoption started in late 1990s. Today, most near-surface applications make use of multistation approaches. They are often identified with the acronym MASW (multistation analysis of surface waves), which was initially introduced by researchers at the Kansas Geological Survey (Park et al. 1999).

Multistation approaches employ a linear array of geophones in line with an active source. Several techniques can be used to process the data, the most common being transform-based approaches. Field data collected in the time–space domain are transformed into a domain (e.g., the frequency–wavenumber domain) where the phase velocities associated with different frequencies are easily chosen by picking the spectral maxima (see Section 4.6).

1.4.3.4 Microtremor surveys

The SASW and MASW methods are active-source tests in which the wave field is generated by a seismic source acting on the ground surface. As mentioned previously, microtremors and ambient vibrations generated by natural events or human activities contain low-frequency energy to characterize the subsurface at depth without resorting to large, heavy active sources. Microtremor surveys, also called passive surface wave tests, record ambient noise using an array of geophones and then analyze these data to extract the experimental dispersion curve. The data are typically collected using a 2D array of receivers because the position of the sources is not known a-priori. Several processing techniques are available, with the most prevalent being the frequency-domain beamforming (Lacoss et al. 1969) and the spatial autocorrelation (Aki 1957) techniques.

The refraction microtremors (ReMi) approach (Louie 2001) is a variant of a microtremor survey using a simple linear array of receivers, although the method has nothing in common with refraction surveys apart from the receiver configuration. In this case, the dispersion curve is estimated using the much simpler approach adopted for active surface wave tests, relying on the assumption that sources of ambient vibrations are uniformly distributed in space. Although this approach is very simple and fast, it can lead to significant

overestimation of the shear wave velocity profile if the background noise is traveling along a preferential direction not in line with the receiver array (Strobbia and Cassiani 2011).

For a given source, the amplitude spectrum of surface waves is a function of the site velocity structure and the source spectrum. However, inverting the amplitude spectrum is not practical because deconvolving the effect of the source is not straightforward due to the fact that the effective source radiation pattern in the far-field is not known even with controlled sources. The use of the ratio between the vertical and horizontal-radial component of the surface waves solves the issues related to the estimation of the source. The polarization, or ellipticity, of the surface wave as a function of the frequency can be inverted to estimate a velocity profile. This approach, used in multichannel, multicomponent data (Muyzert 2007a, 2007b), is also the basis of single-station, multicomponent passive surveys. The average horizontal-to-vertical spectral ratio (HVSR or simply H/V) of passive data is measured and interpreted to estimate the site natural frequency or is inverted (Fäh et al. 2003).

1.5 CHALLENGES OF SURFACE WAVE METHODS

As described earlier, surface wave tests are indirect methods in which the soil properties are estimated on the basis of observed field data by assuming a soil model and inferring its parameters via an inverse problem. The interpretation of experimental data is not straightforward because it is based on a series of complex processing steps. Some assumptions are implicit in the interpretation procedure and their relevance and implications should be evaluated with care. In the following, some specific aspects that play a major role in the interpretation are discussed.

1.5.1 Sampling in space: Apparent phase velocity (mode superposition)

Surface wave tests are based on the geometric dispersion of Rayleigh waves in vertically heterogeneous media. From a mathematical point of view, the propagation of surface waves generated by a point source in vertically heterogeneous media can be represented as the superposition of free Rayleigh modes (see Section 2.4). A dispersion curve is associated with each free Rayleigh mode; thus, the representation shown in Figure 1.3 was simplified in the sense that only a single mode of propagation was displayed.

An ideal experimental survey would be able to extract and identify the dispersion curve for each mode. These data could be used for a robust and effective inversion process (e.g., Gabriels et al. 1987). In reality, the number

of receivers is limited and other factors, such as wave attenuation caused by dissipative behavior of soils, make this ideal survey unachievable (see Chapter 3). Using a limited number of receivers, a single, *apparent* dispersion curve is typically extracted from the field data.

In some situations, the fundamental mode assumes a predominant role in the propagation of Rayleigh waves, meaning that higher modes are not excited by the applied source or, if they are, the energy associated with them is negligible. In such situations, the fundamental mode will be easily extracted via signal processing and the inversion process can be simplified.

Whenever higher modes play an important role in the propagation, the processing of experimental data recovers an apparent or effective phase velocity that is influenced by several modes (see Figure 1.9). In such situations, the inversion process cannot be based on the free Rayleigh modes and the modal superposition must be accounted for in the solution of the forward problem (see Section 2.4). In this case, the inversion becomes more onerous and the stability of automated search algorithms is adversely affected. An alternative strategy can be based on more sophisticated processing techniques allowing for filtering different mode components before the extraction of the experimental dispersion curve.

A fundamental mode approach is more easily implemented and less computationally intensive so that, whenever appropriate, it is the preferred choice. In general, it is not possible to define a rigorous rule to assess the relevance of higher modes. The fundamental mode is typically predominant when the shear wave velocity profile increases gradually with depth. Higher modes

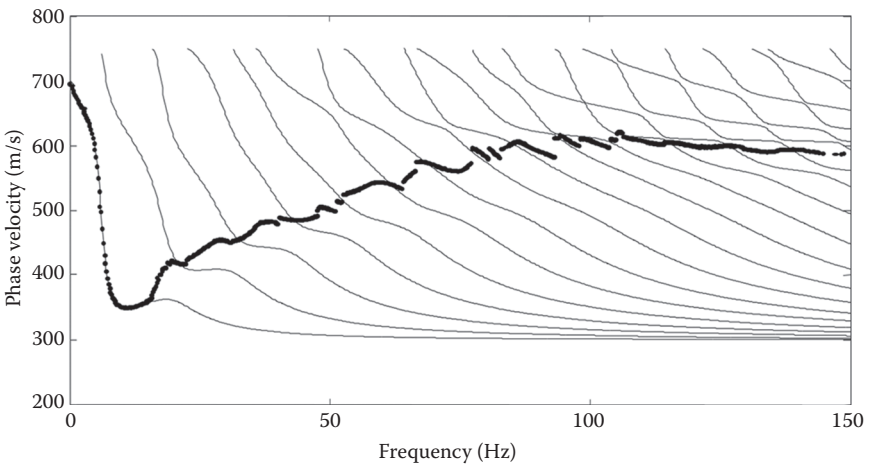

Figure 1.9 Example of modal superposition: an apparent curve results from the superposition of multiple modes at an inversely dispersive site.

become prevalent in the high-frequency range in the presence of a stiff, top layer (e.g., a stiff soil crust or a pavement system as in Figure 1.9), whereas they can be important in the low-frequency band if an abrupt increase in stiffness is present (e.g., shallow bedrock overlain by soft soils).

1.5.2 Near-field effects

A common assumption in surface wave tests is that the wave field comprises only plane surface wave components. In reality, for most applications, the waves are generated by a point source on the ground surface and the receivers are placed in the vicinity of the source so that so-called near-field effects come to play a role. Near-field effects are mainly due to the following:

1. Body wave interference
2. Cylindrical wave front of Rayleigh waves

The wave field in the vicinity of the source is typically quite complex because body wave components and surface wave components are not well separated at this stage. Body wave components attenuate at a much faster rate than surface waves. This is because the wave front is spherical for body waves and cylindrical for surface waves, inducing a different spreading of energy in space. As a consequence, the amplitude of particle motion associated with body waves attenuates proportionally to the inverse of the distance from the source, whereas the particle motion of surface waves attenuates proportionally to the inverse of the square root of the distance from the source. As the wave train travels away from the source, the relative contribution of body wave components decreases so that in the far-field it is acceptable to neglect the influence of body waves.

The second aspect is also related to the spreading of surface waves. In most techniques adopted for determining the experimental dispersion curve, it is assumed that the wave front is plane. Again this assumption is valid only in the far-field, whereas in the near-field the cylindrical shape of the wave front cannot be neglected and accurate analysis requires the use of transforms employing cylindrical coordinates (Zywicki and Rix 2005).

Near-field effects have been extensively studied for the two-station SASW method. The body wave components exert a larger influence on the closest receiver, and as a consequence the overall estimate of the phase velocity is biased. This is one of the major reasons for using several testing configurations in the SASW test as discussed in Section 4.3. Numerical simulations of the complete wave field generated by a point source have shown that near-field effects can affect the dispersion curve estimate if the closest receiver is placed within a certain critical distance from the source. This critical distance can range from one-half to two wavelengths depending on the soil profile (Sànchez-Salinero 1987; Tokimatsu 1995).

Near-field effects can be taken into account by using forward models able to reproduce the complete wave field, but this inevitably leads to computationally intensive inversion methods (e.g., Ganji et al. 1998).

The adverse influence of near-field effects in multistation approaches is alleviated because longer receiver arrays are used so that a larger number of receivers are in the far-field. Moreover, the processing techniques can separate body and surface wave components and identify the portion of data that can be used. Low-frequency components exhibit more near-field effects, requiring the adoption of adequate countermeasures (Strobbia and Foti 2006). A typical strategy is to increase the offset of the first receiver of the array with respect to the source position (Yoon and Rix 2009). Although effective in mitigating near-field effects, this approach also causes the loss of high-frequency components because of internal energy dissipation in soils and hence a loss of resolution in the shallow part of the profile.

1.5.3 Model errors

As discussed previously, surface wave testing is based on the solution of a model parameter identification process. The choice of the reference model is a critical aspect because the experimental data will be fit to the model regardless of whether it is appropriate for the specific site. The most common choice for the model is a stack of homogenous linear elastic layers (Figure 1.10). This is a 1D model that neglects lateral variations in soil properties. Clearly, this is only an approximation of reality, and the more the real conditions differ from this model, the less accurate will be the shear wave velocity profile. The reasons for choosing such a simplified model are: (1) the solution of the forward problem can be obtained very efficiently using a variety of algorithms such as propagator matrix (Thomson 1950; Haskell 1953; Gilbert and Backus 1966)

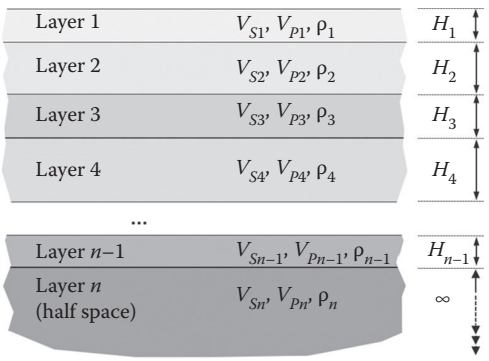

Figure 1.10 Subsurface model commonly adopted for inversion: an 1D stack of linear elastic, homogeneous layers.

or stiffness matrix (Kausel and Roesset 1981) algorithms and (2) the number of unknowns is limited. Both features are helpful when formulating and solving the inverse problem because they reduce the computational effort and reduce the ill-posedness of the inverse problem, respectively.

The number of unknowns may be further reduced by selecting a priori the parameters to be estimated in the inversion process. For the layered model, four parameters are needed to fully characterize each layer (Figure 1.10): density ρ, thickness h, and two elastic constants (e.g., shear modulus G and Poisson's ratio ν). Often, the elastic parameters are defined in terms of the P-wave and S-wave velocities for each layer. On the basis of sensitivity analyses (Nazarian 1984), two of these parameters are typically fixed a priori for each layer, that is, the density and Poisson's ratio (or a given ratio between P-wave and S-wave velocities) of each layer. This strategy reduces the number of unknowns from $4n-1$ (because the thickness of the half-space is not defined) to $2n-1$, where n is the number of layers including the half-space. Sometimes the layer thicknesses are also fixed on the basis of a priori information (e.g., borehole logs) or when a large number of layers are introduced to adequately reproduce the variation of S-wave velocity with depth.

Special care must be used for the selection of the Poisson's ratio. The presence of pore water in a saturated, porous medium is reflected in a continuum mechanics model by a variation of the Poisson's ratio because soil behavior is undrained under dynamic loads. Hence the appropriate value of Poisson's ratio, or equivalently the P-wave velocity, should be assumed taking into account the presence and position of the water table (see Section 7.1.2). Above the water table, the P-wave velocity is governed by the stiffness of the soil skeleton and the Poisson's ratio is typically between 0.2 and 0.3, while in a saturated, porous medium, the velocity of P-wave is governed by the bulk compressibility of the water (for soils) and the Poisson's ratio is close to 0.5.

Model errors arise whenever the model is not appropriate for the real subsurface conditions. Problems may be caused by lateral inhomogeneities or inclined bedding in the strata. Surface wave testing based on a 1D model is not appropriate if these features are marked. It can be useful to perform measurements along perpendicular lines to evaluate the differences in the measured dispersion curves. This strategy may allow an assessment of the hypothesis of plane and parallel layers. An approach for the evaluation of lateral variation is based on tests with moving receiver arrays or moving spatial windows in datasets with a large number of receivers. It is then possible to estimate a series of shear wave velocity profiles for adjacent locations, which provides a pseudo-2D or pseudo-3D model (Socco et al. 2009).

Another situation in which the usual layered model is not appropriate is when the stiffness profile is not characterized by sharp contrasts in properties between adjacent layers but rather by smooth variations with depth. This is the case for homogenous soils in which the variation of stiffness with depth is associated with the increase of confining stresses.

In such conditions, the choice of a different model with smooth variations of properties can be more appropriate (Rix and Lai 2013). An alternative strategy is to use a large number of layers with a constraint in the inversion requiring smoothness of the solution (see Chapter 6). The final solution will in this case resemble the continuous variation with a steplike gradual increase of stiffness in the layered model.

1.5.4 Resolution and depth of investigation

It is intuitive that the resolution of surface wave tests decreases with depth. Thin layers are well resolved when they are close to the ground surface, whereas at great depth the resolution is limited and only large changes can be detected. Indeed, small perturbations of the stiffness profile at great depth (e.g., thin layers of different material) have a modest effect on the dispersion curve for the site. This is illustrated in Figure 1.11, where the percent change of the fundamental-mode phase velocity dispersion curve due to a 10% variation in the velocity of different layers is shown. The magnitude of the change in the phase velocity and the affected frequency range decreases as the depth of the layer increases. The reduction of the sensitivity with depth results in a loss of resolution or in the ability to identify the properties of thin layers. Thus, these features cannot be accurately resolved, especially in the presence of uncertainty in the experimental data.

The distribution of frequencies contained in the experimental dispersion curve also plays a role. It is instructive to plot the experimental dispersion curve in terms of phase velocity versus wavelength rather than in terms of phase velocity versus frequency (Figure 1.12). If the dispersion curve is sampled at evenly spaced frequencies over a particular interval, this will result in a concentration of data points at shorter wavelengths within that interval. Conversely, a wide range of wavelengths (zone "a" in Figure 1.12) is represented by a narrow range of low frequencies. Because the depth of penetration is closely related to wavelength, the consequence is that a relatively large amount of information is available for the portion of the soil profile close to the ground surface, whereas few data points are available to assess the medium properties at depth. The distribution of the information in the experimental dispersion curve is reflected in the solution of the inversion problem. Most inversion algorithms search for the best fitting profile by minimizing the distance between theoretical and experimental dispersion curves. The uneven distribution of information means that the agreement in the short wavelength range, where most data points are, will be most influential in the choice of the final profile.

Several strategies can be devised to overcome this problem. One possibility is to sample or resample the dispersion curve during post-processing with a nonuniform distribution of frequencies (e.g., with an exponential distribution).

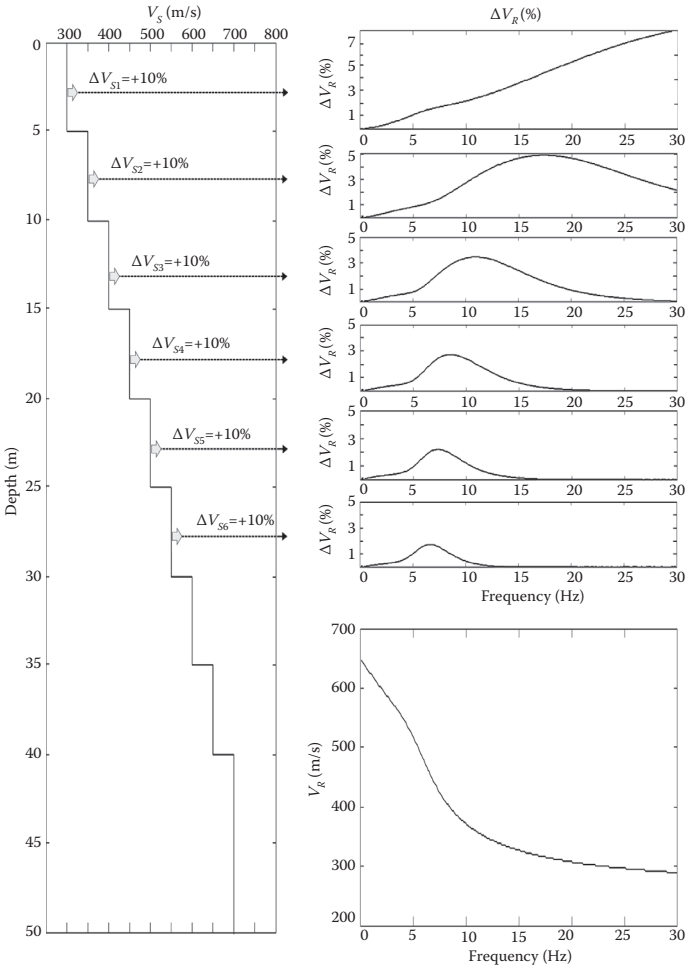

Figure 1.11 Sensitivity of the dispersion curve to changes in the shear wave velocity of layers at different depths.

Another possibility is to assess the fit between the theoretical and experimental dispersion curves in the wavelength domain, but this poses practical difficulties because the theoretical and experimental data are not sampled at the same wavelength.

A third factor that causes loss of resolution with increase depth is random error in the experimental dispersion curve. These random errors increase with decreasing frequency (Lai et al. 2005).

The loss of resolution with depth is ultimately reflected in the uncertainties associated with the shear wave velocity profile estimated from surface

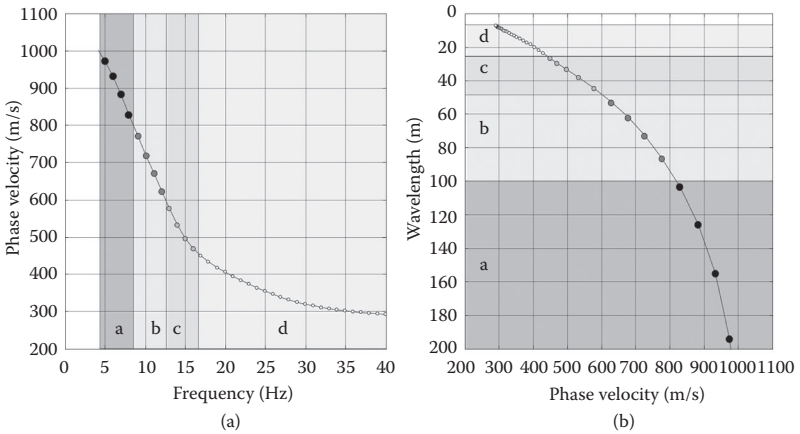

Figure 1.12 Different representations of the Rayleigh wave dispersion curve: (a) phase velocity versus frequency; (b) phase velocity versus wavelength.

wave measurements (see Chapter 6). Such uncertainties are typically quite low close to the ground surface, but they tend to increase with depth.

The other important aspect is related to depth of penetration, that is, the depth of the soil profile that can be characterized using surface wave testing. The answer is related to the available experimental information in the low-frequency range. A preliminary estimate of the maximum depth that can be resolved is obtained by considering the point in the experimental dispersion curve with the lowest frequency, which is usually associated with the longest wavelength. The maximum depth that can be resolved in the inversion is approximately one-half of the longest wavelength because the dispersion curve is insensitive to changes in material properties below this depth. A more precise assessment can be obtained via a sensitivity analysis using the forward solution of Rayleigh wave propagation for the frequency range of the available experimental data.

As a rule of thumb, it is often said that active surface wave methods can resolve the soil profile up to about one-half of the total aperture of the receiver array. The possibility of obtaining information on deep layers is thus limited by logistic aspects and by the necessity of using high-energy sources able to generate surface waves at long offsets. As mentioned previously, passive surface wave tests based on microtremors can be of significant help to extend the depth of penetration.

Although the focus is often on increasing the depth of penetration, the importance of high-frequency information in the solution of the inversion problem should not be overlooked, even if a high level of detail close to the ground surface is not a primary objective of a specific characterization survey. Shallow layers exert an influence on the entire experimental

dispersion curve, and a lack of information for shallow layers is propagated throughout the entire inversion process. In practice, the inability to adequately characterize the shallow portion of the soil deposit will result in a biased model for deeper layers. For this reason, it is important to employ a broad frequency range in the experimental test, and combining active and passive surface wave data is a good strategy in this respect.

1.6 TYPICAL APPLICATIONS

The growing use of surface wave methods in practice and research is demonstrated by the increasing number in scientific papers in the geophysical, engineering, and seismological literature (Socco et al. 2010). In the following sections, a brief review of typical applications is presented with special attention to engineering problems.

1.6.1 Site characterization

The primary use of surface wave tests today is to determine the shear wave velocity profile for site characterization purposes. As mentioned in Section 1.1, shear wave velocity is directly linked to the small-strain shear modulus of soils. The shear wave velocity profile is of primary interest for seismic site response studies and for studies of vibration of foundations and vibration transmission in soils. Other applications are related to the prediction of settlements and to soil–structure interaction.

With respect to the evaluation of seismic site response, it is worth noting the affinity between the model used for the interpretation of surface wave tests and the model adopted for most ground response analyses. The application of equivalent linear elastic methods is often associated with 1D, layered models (e.g., the code SHAKE by Schnabel et al. 1972 and its successors). This affinity is also particularly important in the light of equivalence problems, which arise because of the nonuniqueness of the solution in inverse problems. Indeed, profiles that are equivalent in terms of Rayleigh wave propagation are typically also equivalent in terms of seismic amplification (Foti et al. 2009).

Many seismic building codes use the weighted average of the shear wave velocity profile in the upper 30 m of the soil profile ($V_{S,30}$) to discriminate classes of soils that have similar site amplification characteristics. Because it is an average of the properties in the upper 30 m, $V_{S,30}$ can be evaluated very efficiently with surface wave methods because it does not require the high degree of accuracy that can be obtained with seismic borehole methods. Indeed, surface wave tests are more economical to perform because they are noninvasive, and they are often the only possible choice when geological or geotechnical considerations do not allow invasive tests to be used.

Figure 1.13 shows a comparison between a surface wave test and invasive methods for a large number of sites in the United States and Italy that exhibit good agreement in the estimate of $V_{S,30}$ for a wide range of soil deposits with different stiffness.

Measurements of surface wave attenuation provide a means to determine the *in situ* material damping ratio profile of near-surface soils. Frequency-dependent surface wave attenuation coefficients can be determined from measurements of Rayleigh wave amplitudes at various offsets (see Chapter 5). The attenuation curve can be subsequently used in an inversion process with an approach similar to the one used for evaluating the shear wave velocity profile. The two inversions can also be performed simultaneously in a coupled analysis of surface wave dispersion and attenuation (see Chapter 6).

In situ measurements provide the opportunity to assess low-strain material damping ratio free from the adverse effects of specimen disturbance. In particular, surface wave techniques offer several advantages compared to borehole methods for damping measurements. First, the presence of the borehole together with poor coupling between borehole and receiver may adversely affect measurements of particle motion amplitudes. Surface wave tests eliminate these problems because the receivers are on the ground surface where proper soil–receiver coupling can be verified. The frequencies used in surface wave tests are also closer to those of interest in earthquake site response analyses than frequencies used in cross-hole or down-hole tests.

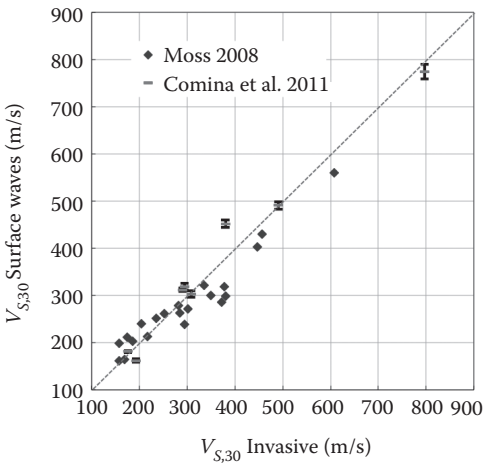

Figure 1.13 Comparison of $V_{S,30}$ determined with invasive tests and surface wave tests at sites in the United States (Moss 2008) and Italy (Comina et al. 2011). For the Italian sites, an estimate of the uncertainty associated with surface wave analysis is provided.

The use of surface wave tests for characterization of municipal solid waste is particularly appealing (Kavazanjian et al. 1996). In such cases, the use of noninvasive methods to obtain the small-strain stiffness and damping is advantageous due to the difficulties and danger associated with the collection of samples and to the realization of boreholes and intrusive testing. Moreover, the advantage of measuring average properties of the waste material is in this case emphasized because of the large variability in particle dimensions (Haegeman and Van Impe 1998; Rix et al. 1998).

1.6.2 Soil improvement

Surface wave tests can be used as a monitoring tool to check the effectiveness of soil improvement techniques at a site by repeated realizations of the test. The comparison of the experimental dispersion curve at different times gives a preliminary and rapid evaluation of the evolution of ground properties, whereas detailed information concerning the effectiveness at different depths is provided by the shear wave velocity profiles obtained via inversion. Successful applications are reported in the literature for monitoring compaction for liquefaction mitigation in sands (Andrus et al. 1998) and for effects of preloading in clays (Raptakis 2012). The relationship between shear wave velocity and soil density can also be used for quality assurance of soil compaction in embankments and fills (Kim et al. 2001).

Similar to the case of soil improvement, applications of surface wave testing to study permafrost evolution in cold regions are reported in the literature. The repetition of the test at different times of the year can yield information about the seasonal variations of soil stiffness caused by freezing cycles (Alkire 1992). The investigation of the permafrost zone is particularly significant for applications related to slope stability and foundation systems.

1.6.3 Nondestructive testing of pavements

The characterization of pavements is a straightforward application of surface wave methods because they are typically horizontally layered media that conform well to the model used in interpretation. Indeed, the early applications of the two-station SASW method were mainly related to pavement systems (Heisey et al. 1982; Nazarian 1984). The primary objectives are quality assurance during construction and evaluating the degradation of the pavement system during its life cycle to optimize maintenance activities.

High-frequency information is necessary because of the scale of the application, and this is usually accomplished from an experimental point of view using accelerometers rather than geophones as receivers. The inversely dispersive nature of pavements (i.e., stiffness decreasing with depth) complicates the inversion procedure (Al-Hunaidi 1998; Ryden et al. 2004; Ryden and Park 2006). Apparent or effective experimental dispersion curves are

typically obtained in such situations, and the contribution of higher modes must be appropriately accounted for (see Section 1.5.1). On the contrary, the geometry of the system is often known a priori, which reduces the number of unknown parameters and allows the inversion to focus on obtaining the elastic moduli of the pavement layers.

1.6.4 Offshore and near-shore site characterization

Applications of surface wave testing underwater are based on the generation and detection of Scholte waves, which are the equivalent of Rayleigh waves on the seafloor (i.e., the interface between a fluid layer and elastic half-space). The geometric dispersion of Scholte waves is influenced by the thickness of the fluid layer. The dispersion curve for Scholte waves approaches the Rayleigh wave dispersion curve as the thickness of the fluid layer diminishes. Proper interpretation of marine tests requires the implementation of the forward problem for Scholte wave propagation by modeling a fluid layer above linear elastic layers.

Theoretical and experimental studies have been carried out by researchers at the University of Texas at Austin to assess the possibility of applying the two-station SASW method on the seafloor (Manesh 1991; Luke 1994). In their application, the source and receiver were manually placed on the seafloor (Luke and Stokoe 1998). Obviously this strategy is only viable for shallow water applications.

Many applications are reported in the literature using multistation data acquired from seismic vessels typically used for deep exploration for the oil and gas industry. Sources are typically towed by vessels, and receivers can be embedded in a towed streamer or deployed at the sea bottom via cables or independent stations.

1.6.5 Near-surface characterization in seismic exploration

In seismic reflection surveying, shallow subsurface deposits are responsible for two of the main challenges associated with land applications: perturbations and distortions induced by the near-surface and coherent noise consisting of high-energy, long-duration, near-surface modes.

In large-scale seismic exploration, the shallow subsurface often has large lateral variability. Heterogeneities in the near-surface can be responsible for large distortions of the body-wave wave field, and can impact the final images of the exploration targets and the extracted seismic attributes. In seismic reflection surveying, P-waves must travel twice through this heterogeneous, usually low-velocity portion of the subsurface, downward from sources at the surface to the deep reflectors and then upward to the receivers. The two-way travel time contains the contribution of the near-surface on

the source and on the receiver side. Traditionally, the correction of the near-surface perturbations is performed with static corrections derived from near-surface models obtained via refraction data.

The near-surface is also the portion of the subsurface in which the surface waves propagate. The complexity of the surface wave waveform, with dispersion and multiple modes, in fact depends directly on the velocity variations within the subsoil. Surface waves have been traditionally considered as coherent noise to be removed, and they are conventionally referred to as "ground-roll." Mitigation of these effects is achieved with a combination of field acquisition practices, such as the use of source and receiver arrays, and with digital filters during data processing.

Recently, progress in seismic data acquisition and data processing has changed this conventional perspective on surface waves (Strobbia et al. 2011). Broadband, point-receiver data with three-dimensional (3D) geometries offer an optimal sampling of the surface waves. Low-frequency sources and receivers allow the extension of the frequency band down to 1 Hz and, therefore, the acquisition of long wavelengths needed for a meaningful investigation depth. At the same time, finely spaced point receivers provide a dense spatial sampling, enabling the observation of short wavelengths needed for the resolution in the shallow subsurface and for stable surface wave inversion.

Without substantial differences in the general principles, surface wave methods in reflection surveying benefit from the optimal sampling and extreme redundancy of modern 3D geometries. The sampling is not 1D and evenly spaced, and the processing requires advanced approaches, but the number of traces that can be used is very large. With a local array of receivers, thousands of shots can be used. Moreover, powerful land seismic sources generate high-energy surface waves and can create a broad spectrum, spanning two orders of magnitude from 1 to 100 Hz. For these reasons, the analysis can produce accurate, high-resolution data describing the surface wave propagation properties at the scale of interest. The results can yield velocity models of the near-surface over large areas, revealing geological complexity such as structural features, lineaments, and lithological boundaries. Figure 1.14 shows a slice of a dispersion volume, which is the equivalent of a dispersion curve for a 3D survey. It represents the phase velocity as a function of the wavelength (or the frequency) for a specific surface location.

The velocity models obtained from surface wave inversion can be used to compute perturbation corrections or merged with deeper velocity models for depth imaging. The detailed identification of the propagation properties also allows the generation of synthetic model traces of the surface waves, which can be subtracted or adaptively subtracted from the raw data. This model-based approach to the surface wave removal has the advantage of avoiding large, multichannel filters, which can distort useful signal features. Moreover, the model-based approach allows dealing with aliased surface waves.

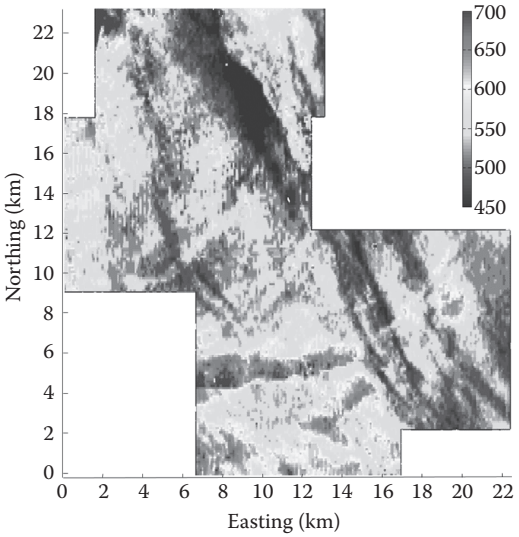

Figure 1.14 Slice of a 3D dispersion volume representing the phase velocity at constant wavelength over a large area.

1.6.6 Anomaly detection

Although estimating the soil stiffness and/or damping profile is the most common objective of surface wave testing, other applications have been proposed that are related to the detection of voids and inclusions in soil deposits. Numerical simulations (typically using the finite element method) and lab-scale tests have been conducted to identify perturbations of the wave field caused by anomalies (Gucunski et al. 1996, 1998; Ganji et al. 1997; Nasseri-Moghaddam et al. 2005). The identification of the anomaly is typically based on the recognition of particular features in the signals (using several signal analysis techniques) rather than on the solutions of a formal inverse problem. Some applications based on pseudo-2D (two-dimensional) modeling are also reported in the literature (Xu et al. 2008).

In large-scale surveying, the spatial distribution of near-surface velocity can be used to identify faults and structural elements (in the geologic meaning). This is useful for the geological modeling and also for drilling safety in hydrocarbon exploration.

1.7 ADVANTAGES AND LIMITATIONS

This last section is devoted to briefly outlining advantages and limitations of surface wave tests. As with any geophysical method, it is important to understand what can and what cannot be achieved.

The first limitation of standard surface wave methods is related to the underlying geophysical model: the assumption of a laterally homogeneous model affects the processing and the inversion. Typically, a stack of linear elastic layers is used as a forward model. A single surface wave test cannot identify lateral variation, and the final result is biased if the soil deposit does not resemble reasonably a 1D, layered medium at the scale of the test. Most of the approaches proposed for the construction of 2D and 3D models from surface wave data are still based on a series of 1D analyses, and as such they should be used with particular care and with a clear understanding of the actual procedure (see Chapter 6). True 2D or 3D testing is possible, but it requires more advanced processing and inversion strategies.

Because inverse problems are mathematically ill-posed, the nonuniqueness of the solution is a limitation. Several profiles that yield theoretical dispersion curves that have a similar distance from the experimental dispersion curve can be identified. This problem is well known as equivalence in geophysical tests based on inverse problems. The implication is a certain degree of uncertainty in the final shear wave velocity profile. For example, surface wave tests are likely not the first choice when the objective is the exact location of an interface between different layers.

The assessment of the uncertainty of the final result is important to evaluate the reliability of the solution. The resolution of the shear wave velocity profile that can be obtained with the surface wave method decreases with depth; thin layers can be resolved if they are close enough to the ground surface, but they are not detected if they are at depth.

Despite their limitations, surface wave methods provide an excellent tool for soil characterization if the overall behavior of the medium is the objective. One important advantage derives from the noninvasive nature of the test: source and receivers are located on the ground surface, which eliminates the need for boreholes. For this reason, they are cost and time effective and can be performed where it is not advisable to invade the medium (e.g., waste materials).

Compared to other seismic techniques, surface wave methods are robust with respect to data acquisition. Compared to shear wave refraction and reflection methods, the acquisition of high-quality surface wave data is easier. Due to the high-energy nature of the surface waves, acquisition is possible even in noisy environments (e.g., urban areas or industrial sites). Other seismic tests based on the evaluation of first arrivals and travel times are more difficult to interpret in the presence of background noise. In surface wave tests, background noise can even be used as a source of information using microtremors surveys.

Surface wave methods are also robust with respect to the model complexity and do not have limitations related to the stratigraphy of a site; they are able to characterize the medium irrespective of the sequence of stiffer (faster) and softer (slower) layers. They work in the case of large contrasts

of velocity and in the case of smooth profiles. Refraction-based techniques, also popular in near-surface applications, have intrinsic limitations (velocity inversions, hidden layers) that can result in ambiguities. The joint inversion or interpretation of refracted and surface waves offers many synergies (Foti et al. 2003).

From a geotechnical engineering perspective, large volumes of soils are tested and the test results reflect the overall dynamic behavior of the soil deposit. The degree of accuracy obtained with a surface wave test is typically in line with the assumptions and the simplifications adopted in geotechnical design. Moreover, the 1D model used for the interpretation is also common in many engineering analysis and design procedures (e.g., the code SHAKE for seismic site response analysis). From an exploration geophysics viewpoint, surface waves can be inverted for the near-surface characterization and then removed for the reflection processing. Thus, surface wave tests can be used for evaluating subsurface profiles at many different engineering scales, ranging from pavements to deep soil profiles. They are a flexible tool to explore the subsurface conditions.